Illustration 11.1

If $F(x) = x^2 + 5$ then

$F'(x) = 2x$.

Thus if $f(x)$ is defined by

$f(x) = 2x$, then

we say that $f(x)$ is the derivative of $F(x)$ and that $F(x)$ is an antiderivative of $f(x)$

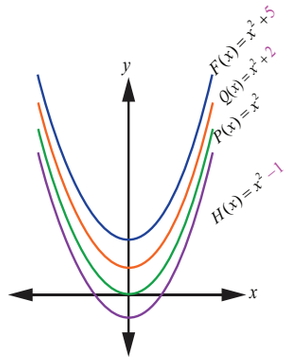


Fig. 11.5

Consider the following table

$F(x)$	$F'(x) = f(x)$	Antiderivative of $f(x) = 2x$
$P(x) = x^2 + 0$ $Q(x) = x^2 + 2$ $H(x) = x^2 - 1$	$P'(x) = 2x$ $Q'(x) = 2x$ $H'(x) = 2x$ } $f(x) = 2x$	$F(x) = x^2 + ?$

We can see that the derivative of $F(x), P(x), Q(x)$ and $H(x)$ is $f(x)$, but in reverse the antiderivatives of $f(x) = 2x$ is not unique. That is the antiderivatives of $f(x)$ is a family of infinitely many functions.

Theorem 11.1

If $F(x)$ is a particular antiderivative of a function $f(x)$ on an interval *I*, then every antiderivative of $f(x)$ on *I* is given by

$$\int f(x)dx = F(x) + c$$

where c is called an arbitrary constant, and all antiderivatives of $f(x)$ on *I* can be obtained by assigning particular value to c.

The function ***f(x)*** is called **Integrand**.

The variable ***x*** in **dx** is called **variable of integration** or **integrator**.

The process of finding the integral is called **integration** or **antidifferentiation** (Newton-Leibnitz integral).

The peculiar integral sign $\int$ originates in an elongated S (like Σ) which stands for sum.

Often in applications involving differentiation it is desired to find a particular integral antiderivative that satisfies certain conditions called initial condition or boundary conditions.

For instance, if an equation involving $\frac{dy}{dx}$ is given as well as the initial condition that $y = y_1$ when $x = x_1$, then after the set of all antiderivatives is found, if x and y are replaced by x_1 and y_1, a particular value of the arbitrary constant is determined. With this value of c a particular antiderivative is obtained.

Illustration 11.2

Suppose we wish to find the particular antiderivative satisfying the equation

$$\frac{dy}{dx} = 2x$$

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